

Proof. 1. **Assumptions and degenerate cases.** We assume throughout that $X(\omega) \geq 0$ for every $\omega \in \Omega$ and that $a > 0$. If $\mathbb{E}[X] = 0$, then $X = 0$ $\mathbb{P}$ -a.s., whence $\mathbb{P}(X \geq a) = 0$ for all $a > 0$ and the desired inequality holds with equality. Hence we may suppose $\mathbb{E}[X] > 0$.

2. **Indicator of the event $\{X \geq a\}$.** Define the indicator random variable

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & \text{if } X(\omega) \geq a, \\ 0, & \text{if } X(\omega) < a. \end{cases}$$

Clearly $0 \leq \mathbf{1}_{\{X \geq a\}}(\omega) \leq 1$ for all ω .

3. **Pointwise inequality.** For each $\omega \in \Omega$ we have

$$\mathbf{1}_{\{X \geq a\}}(\omega) \leq \frac{X(\omega)}{a}. \quad (1)$$

Indeed, if $X(\omega) \geq a$ then the left-hand side equals 1 and $1 \leq X(\omega)/a$; if $X(\omega) < a$ then the left-hand side is 0 and $0 \leq X(\omega)/a$ because $X(\omega) \geq 0$.

4. **Monotonicity of expectation.** Since expectation preserves the order of non-negative random variables, (1) yields

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \mathbb{E}\left[\frac{X}{a}\right].$$

5. **Linearity of expectation.** The factor $1/a$ is constant, so

$$\mathbb{E}\left[\frac{X}{a}\right] = \frac{1}{a} \mathbb{E}[X]. \quad (2)$$

6. **Expectation of an indicator equals probability.** For any event $A \in \mathcal{F}$, $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. Taking $A = \{X \geq a\}$ gives

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = \mathbb{P}(X \geq a). \quad (3)$$

7. **Conclusion.** Combining (2) and (3) with the inequality of step 4 we obtain

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

which is Markov's inequality.

8. **Equality case.** Equality throughout the chain occurs precisely when equality holds in the pointwise estimate (1) for $\mathbb{P}$ -almost every ω . Thus we must have

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \frac{X(\omega)}{a} \quad \text{a.s.}$$

which forces

$$X(\omega) = 0 \text{ whenever } X(\omega) < a, \quad X(\omega) = a \text{ whenever } X(\omega) \geq a.$$

Hence X takes only the two values 0 and a (the trivial case $X \equiv 0$ being included). Conversely, if X assumes only these two values, then $\mathbb{P}(X \geq a) = \mathbb{P}(X = a)$ and $\mathbb{E}[X] = a \mathbb{P}(X = a)$, so the bound is attained. $\square$